

CORE COMPETENCY MATRIX

Avionics & Control: ArduPilot / PX4 Porting, Custom ESP32-S3 Boards, SITL	Autonomy & Middleware: ROS 2 (Humble, Jazzy), Zenoh, Micro-XRCE-DDS, MAVLink
Programming & Systems: C++17/20, Python, C, Linux Shell, Docker	Perception & Mapping: ORB-SLAM3, OpenCV, Octomap Voxel, Gazebo Harmonic
Aerospace Design: OpenVSP CAD, PrePoMax FEA, OpenFOAM CFD, Pulsejet Prototypes	Quality Inspections: NDT Level II Certification, Casting & Weld Auditing, PMI

PROFESSIONAL EXPERIENCE

UAV Systems & Flight Operations Engineer | Novatech Robo Pvt Ltd

JUN 2024 — PRESENT

- Industrial Operations:** Commanded rigorous flight-testing, sensor calibration (IMUs, barometers, transceivers), and field deployments for a fleet of 20+ custom commercial and visual-mapping UAV platforms, achieving a 100% mission safety and aircraft recovery record.
- Telemetry & Predictive Health:** Engineered automated Python-based telemetry parsing pipelines to extract and audit real-time flight logs, analyzing IMU high-frequency vibrations, EKF3 residuals, and voltage transients to reduce structural vibration bounds by 25%.
- Hardware-in-the-Loop Integration:** Successfully assembled and wired custom ESP32-S3 control stacks, level-shifting electronics, and ducted mapping cameras, authoring strict pre-flight checklists and offboard ROS 2 path-planning behaviors.
- Compliance & Safety Leadership:** Authored and directed corporate operational SOPs, ensuring strict compliance with national civil aviation frameworks during industrial delivery flight trials.

NDT & Inspection Engineer Intern | Trinity NDT Engineers

FEB 2023 — MAY 2023

- Non-Destructive Testing (NDT):** Conducted rigorous NDT audits across casting components, structural welding joints, and active pipelines.
- Material Quality & PMI:** Executed high-accuracy Positive Material Identification (PMI) scans to verify metallurgy and material composition integrity.
- Standard Compliance:** Verified assembly certifications for high-pressure industrial equipment, including air receiver tanks and structural frames.

FLAGSHIP ENGINEERING PORTFOLIO

- AeroCore-S3 Custom Flight Controller**

(Avionics, Porting & Flight Testing)

Designed and built a custom ESP32-S3 based flight controller from the hardware level. Ported ArduPilot's Hardware Abstraction Layer interface mappings for high-speed SPI sensor communication (BMI270 IMU, BMP581 Baro) and successfully validated EKF3 state estimation during manual and autonomous test flights (live video proof published).
- Project ICARUS: HALE Autonomy & Structural Co-Design**

(OpenVSP, FEA, CFD & PyTorch)

Completed full design and stress boundary certification lifecycle for a 3.0m High-Altitude Long-Endurance (HALE) UAV. Conducted structural stress audits using PrePoMax/CalculiX FEA, simulated aerodynamic coefficients via OpenFOAM CFD, and trained a PyTorch neural geometry optimizer achieving an 11.9% lift efficiency gain.

Project-TRINITY: Decentralized Swarm Engine (C++, ROS 2, Zenoh & RViz)

Developed a lightweight C++ boids swarm coordination library running inside localized ROS 2 namespaces. Integrated Eclipse Zenoh router protocols for peer-to-peer telemetry state exchanges, cutting active network overhead by 40% compared to pure ROS 2 DDS.

EDUCATION & CREDENTIALS

Bachelor of Engineering in Aeronautical Engineering

JUNE 2023

SJC Institute of Technology (VTU) | Aug 2019 — June 2023 | *Thesis Project: Prototype Design of a Valveless Pulsejet Engine*

 **Languages:** English, Hindi, Kannada, Tamil

 **Certifications:** Professional Non-Destructive Testing (NDT) Level II Certification